

BOOSTXL-DRV8305EVM User's Guide

This document is provided with the BOOSTXL-DRV8305EVM customer evaluation module (EVM) as a supplement to the DRV8305 data sheet ([DRV8305 Three Phase Gate Driver With Current Shunt Amplifiers and Voltage Regulator](#)). This user's guide provides details on the setup and hardware implementation of the BoosterPack™ plug-in module.

Contents

1	BOOSTXL-DRV8305EVM	2
	1.1 PCB 3-D Views.....	2
	1.2 PCB 3-D Views.....	3
2	Introduction	4
	2.1 Features.....	4
	2.2 Pinout.....	5
	2.3 Operating Conditions	5
3	Getting Started	6
	3.1 Requirements	6
	3.2 Configuring the LaunchPad	6
	3.3 Connecting the Hardware.....	7
4	Demo Application	7
5	Detailed Hardware Description.....	8
	5.1 DC Bus and Phase Voltage Sense.....	8
	5.2 Low-Side Current Shunt Sense	8
	5.3 BoosterPack GPIO Signals	9
	5.4 DRV8305 Status and Control Registers	9
	5.5 BOOSTXL-DRV8305EVM Schematic	11
	5.6 Hardware Source Files	12

List of Figures

1	PCB Image (From Above)	2
2	BoosterPack Header Signals.....	2
3	3-D Top View.....	3
4	3-D Bottom View.....	3
5	BOOSTXL-DRV8305EVM With LAUNCHXL-F28027F	4
6	BOOSTXL-DRV8305EVM Pinout.....	5
7	LAUNCHXL-F28027F Configuration	6
8	MotorWare™ Software.....	7
9	Voltage Sense.....	8
10	Current Sense.....	8
11	Enable System	9
12	DRV8305 SPI Registers.....	10
13	BOOSTXL-DRV8305EVM Schematic	11

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BoostXL-DRV8305EVM 用户手册

本文件作为 DRV8305 数据手册 ([DRV8305 三相门驱动器 \(含电流分流放大器与电压调节器\)](#)) 的补充资料, 随 boostxl-DRV8305EVM 客户评估模块 (EVM) 一同提供。本用户指南详细介绍了 BoosterPack™ 插件模块的配置与硬件实现方法。

目录

1	BOOSTXL-DRV8305EVM	2
1.1	PCB 三维视图.....	2
1.2	PCB 三维视图.....	3
2	引言.....	4
2.1	功能特性.....	4
2.2	引脚排线.....	5
2.3	运行工况.....	5
3	快速入门.....	6
3.1	需求.....	6
3.2	LaunchPad 配置.....	6
3.3	连接硬件.....	7
4	演示应用程序.....	7
5	详细硬件描述.....	8
5.1	直流母线与相电压检测.....	8
5.2	低侧电流分流检测.....	8
5.3	BoosterPack GPIO 信号.....	9
5.4	DRV8305 状态与控制寄存器.....	9
5.5	boostxl-DRV8305EVM原理图.....	11
5.6	硬件源文件.....	12

图列表

1	PCB原理图 (上方视角)	2
2	BoosterPack 头部信号.....	2
3	三维俯视图.....	3
4	三维底视图.....	3
5	BoostXL-DRV8305EVM (搭配 LaunchXL-F28027F)	4
6	BoostXL-DRV8305EVM 引脚图.....	5
7	launchxl-f28027f 配置.....	6
8	MotorWare™ 软件.....	7
9	电压检测.....	8
10	电流检测.....	8
11	启用系统.....	9
12	DRV8305 SPI 寄存器.....	10
13	boostxl-DRV8305EVM原理图.....	11

商标

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1 BOOSTXL-DRV8305EVM

1.1 PCB 3-D Views

Figure 1 shows the top view of the BOOSTXL-DRV8305EVM board. Figure 2 shows the BoosterPack header signals.

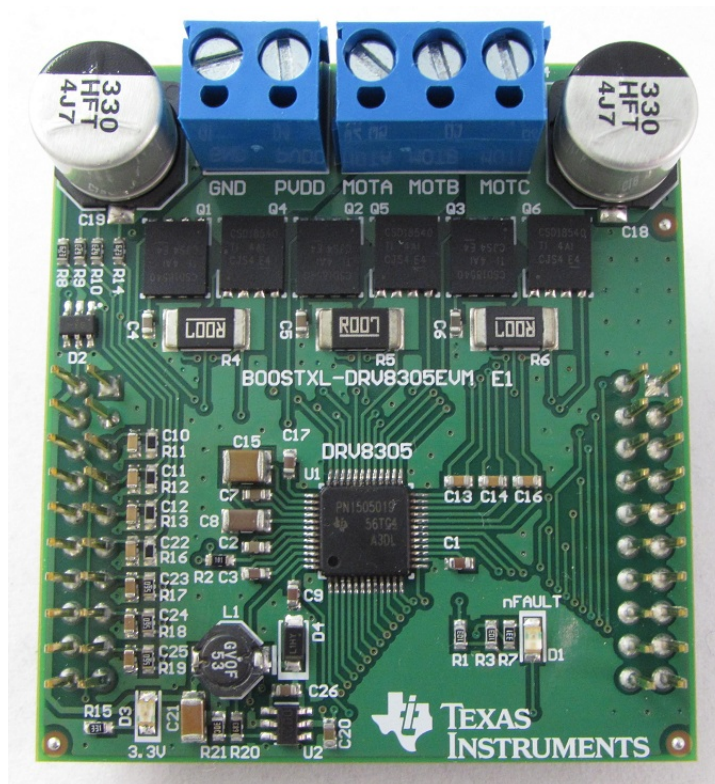


Figure 1. PCB Image (From Above)

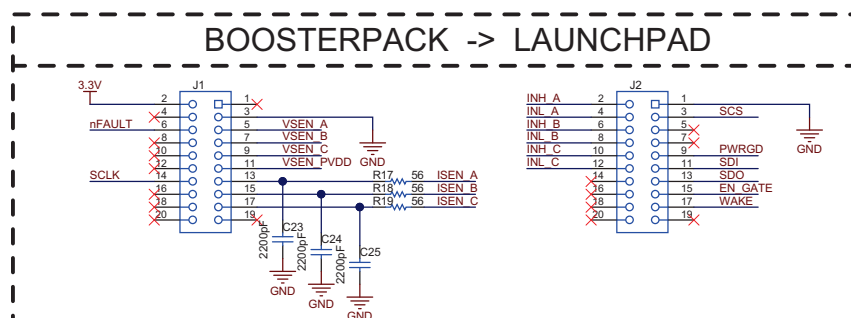


Figure 2. BoosterPack Header Signals

1 BOOSTXL-DRV8305EVM

1.1 PCB 三维视图

图 1 展示了 boostxl-DRV8305EVM 开发板的俯视图。图 2 展示了 BoosterPack 接口信号。

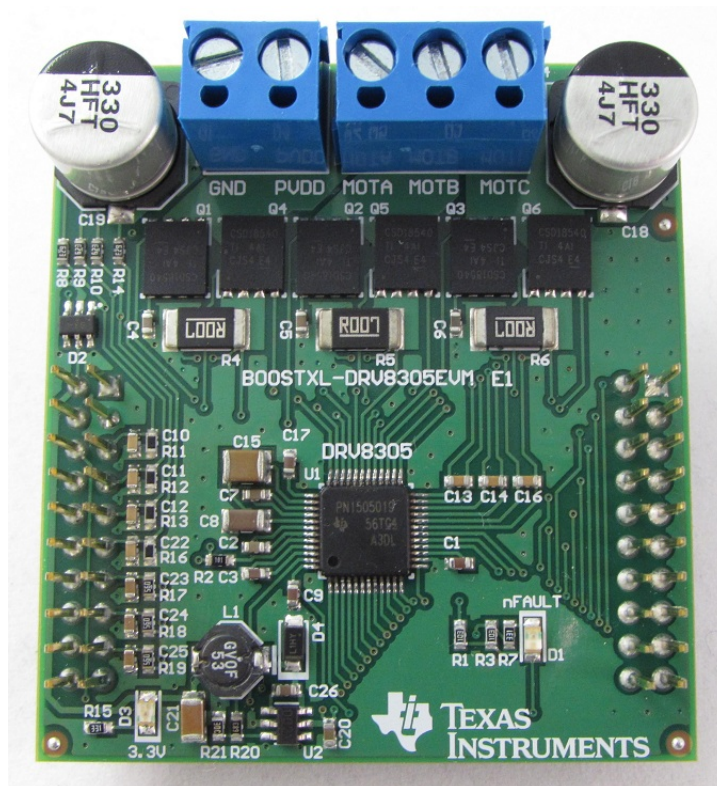


图 1. PCB 图像 (从上方拍摄)

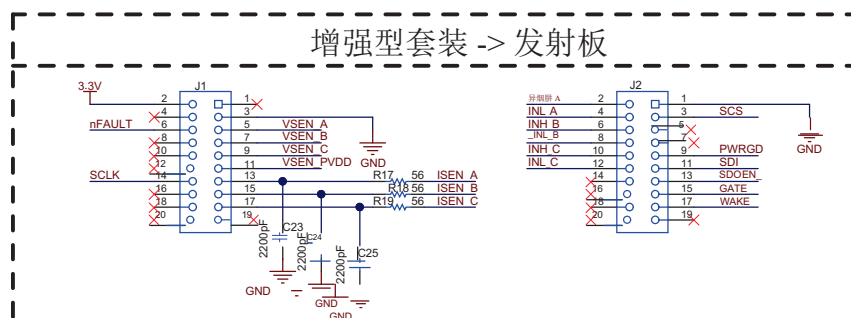


图 2. BoosterPack 头部信号

1.2 PCB 3-D Views

Figure 3 and Figure 4 show the top and bottom three-dimensional PCB views.

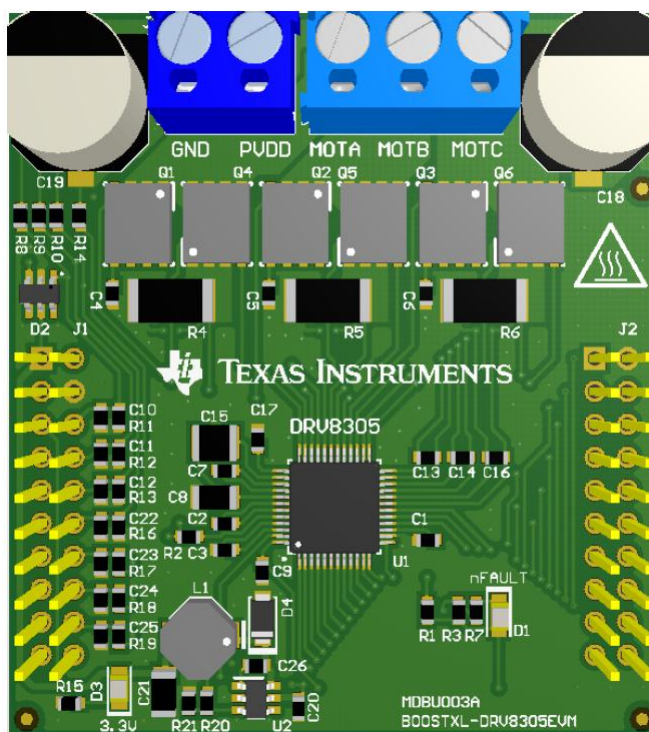


Figure 3. 3-D Top View

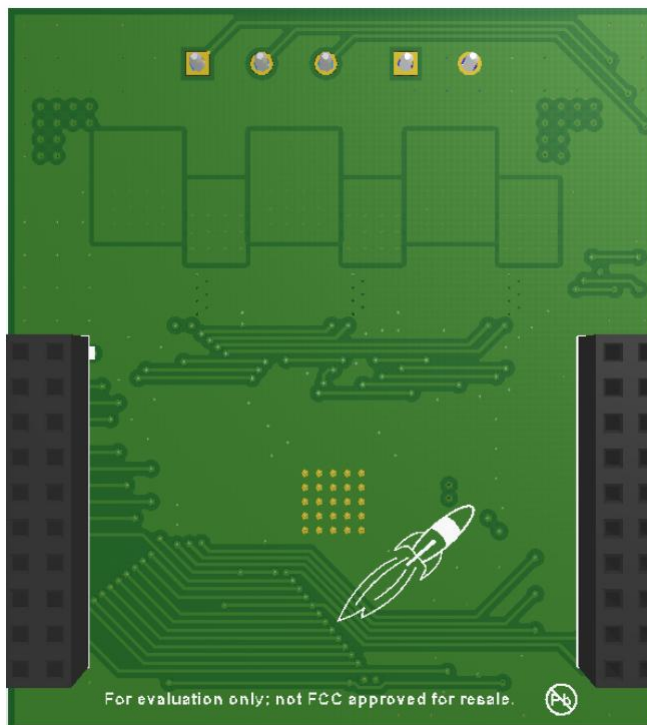


Figure 4. 3-D Bottom View

1.2 PCB 三维视图

图 3 与图 4 分别展示了 PCB 的顶部和底部三维视图。

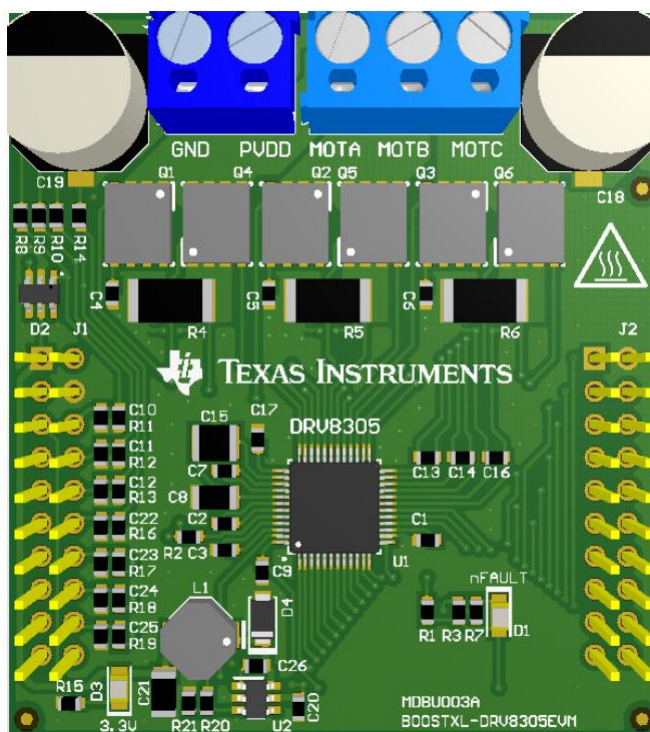


图 3. 三维俯视图

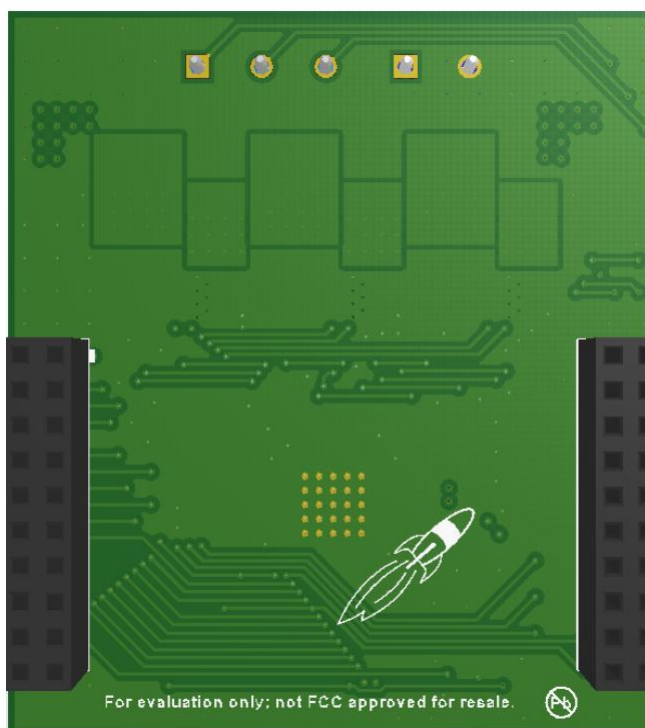


图 4. 三维俯视图

2 Introduction

The BOOSTXL-DRV8305EVM BoosterPack is a complete 3-phase driver stage in order to evaluate motor application with the DRV8305 motor gate driver. It utilizes a compact and modular form factor for ease of use and is designed to dock with compatible TI LaunchPad™ development kits for a complete motor control system.

2.1 Features

The following lists the BOOSTXL-DRV8305EVM features:

- Complete 3-phase drive stage in a compact form factor (2.0 in × 2.2 in)
- Supports 4.4- to 45-V voltage supply and up to 15-A RMS (20-A peak) drive current
- 6x CSD18540Q5B N-Channel NexFET™ Power MOSFETs (1.8 mΩ)
- Individual motor phase and DC bus voltage sense
- Low-side current shunt sense for each half-bridge
- Fully protected drive stage including short circuit, thermal, shoot-through, and undervoltage protection
- LMR16006 wide voltage input, 0.6-A step down buck regulator for MCU supply
- Combine with compatible LaunchPad XL kits to create a complete 3-phase motor control platform
- Optimized for the Piccolo™ LAUNCHXL-F28027F LaunchPad to support the InstaSPIN-FOC™ sensorless motor control solution

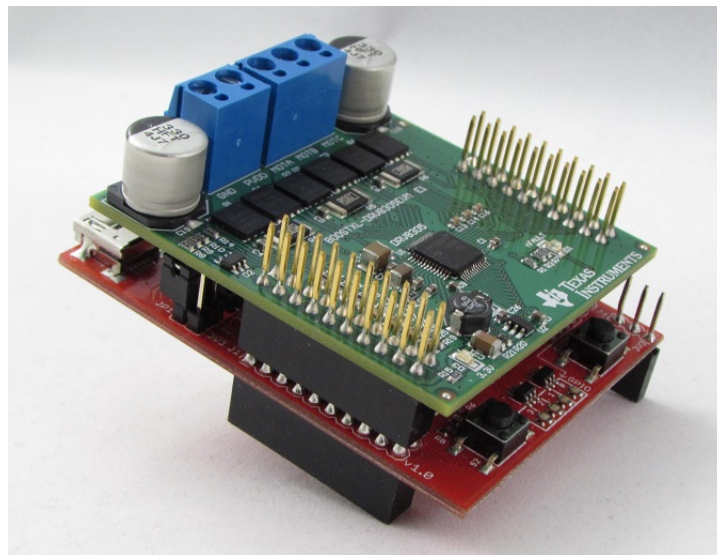


Figure 5. BOOSTXL-DRV8305EVM With LAUNCHXL-F28027F

2 引言

BoostXL-DRV8305EVM BoosterPack 是一款完整的三相驱动级器件，用于配合 DRV8305 电机门极驱动器对电机应用进行评估。该器件采用紧凑、模块化的外形设计，便于使用；同时支持与兼容的 TI LaunchPad™ 开发套件对接，构建完整的电机控制系统。

2.1 功能特性

以下列出了 boostxl-DRV8305EVM 的功能特性：

- 紧凑型三相驱动级（尺寸：2.0 英寸 × 2.2 英寸）
- 支持 4.4 至 45 V 电压输入，以及最高 15 A RMS（20 A 峰值）的驱动电流
- 6 颗 CSD18540Q5B N 沟道 NexFET™ 功率 MOSFET（1.8 mΩ）
- 独立电机相位与直流母线电压检测
- 每个半桥的低侧电流分流检测
- 全保护驱动级，包含短路保护、过热保护、过流保护及欠压保护
- LMR16006：适用于微控制器供电的宽电压输入、0.6 A 降压稳压器
- 与兼容的 LaunchPad XL 开发套件配合使用，构建完整的三相电机控制平台
- 专为 Piccolo™ LaunchXL-F28027F LaunchPad 设计，支持 InstaSPIN-FOC™ 无传感器电机控制解决方案

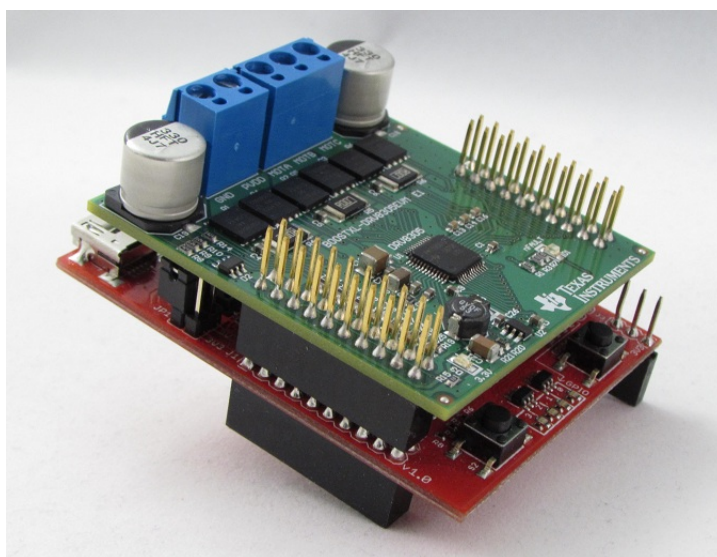


图 5. BoostXL-DRV8305EVM（搭配 LaunchXL-F28027F）

2.2 Pinout

The BOOSTXL-DRV8305EVM brings out a mixture of power, control, and feedback signals to the XL LaunchPad headers.

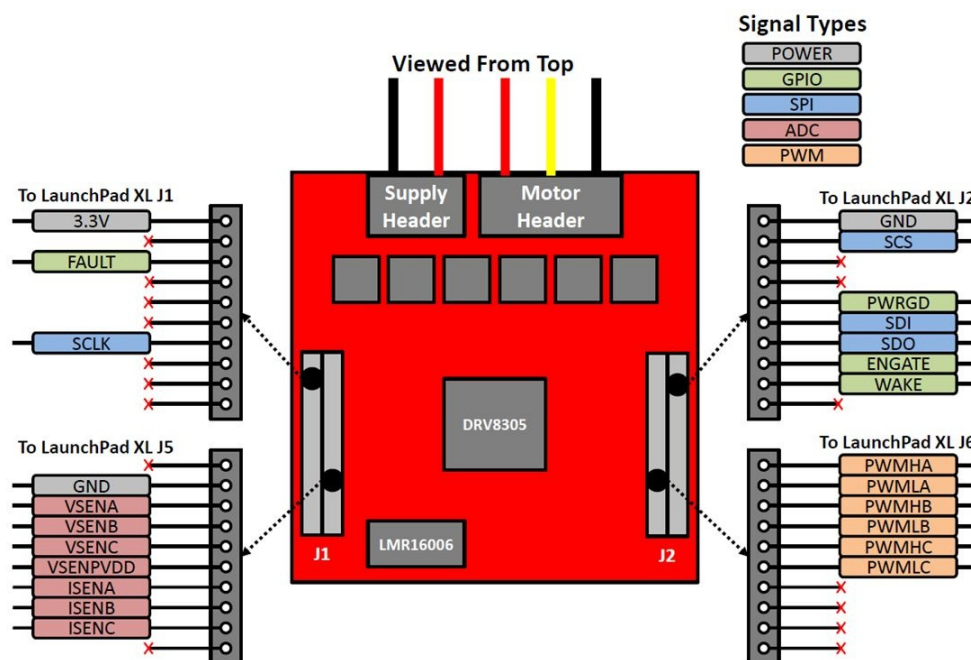


Figure 6. BOOSTXL-DRV8305EVM Pinout

- Terminal block headers for the power supply and motor connections
- Onboard LM16006 step-down buck regulator to provide 3.3-V power to the LaunchPad
- Fault reporting through the nFAULT and PWRGD signals
- SPI to set device configuration, operating parameters, and read out diagnostic information
- Voltage sense for the voltage supply bus and each phase output (scaled for 4.4- to 45-V operation)
- Low-side current shunt sensing on each phase (scaled for 0- to 20-A peak current operation)

2.3 Operating Conditions

Table 1 lists the operating conditions.

Table 1. Operating Conditions

Parameter	Minimum	Maximum	Unit
Operating supply voltage	4.4	45	V
Operating supply current (EN_GATE = Low)		150	mA
Operating temperature	−40	125	°C

2.2 引脚排线

BoostXL-DRV8305EVM 将电源、控制及反馈信号混合输出至 XL LaunchPad 接线端子。

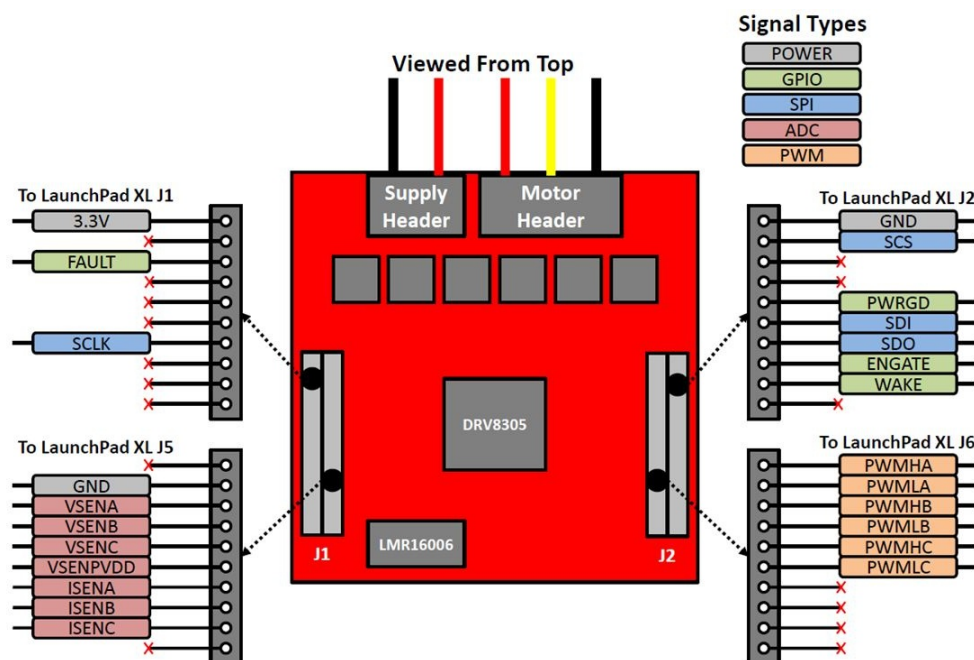


图 6. boostxl.DRV8305EVM 引脚图

- 用于电源及电机连接的端子排接线头
- 搭载LM16006降压稳压器，为LaunchPad提供3.3 V电源
- 通过 nFAULT 和 PWRGD 信号进行故障报告
- SPI接口用于设置设备配置、操作参数及读取诊断信息
- 用于电压供电总线及各相输出端的电压检测（针对 4.4–45 V 工作范围进行比例缩放）
- 每相低侧电流分流检测（针对0–20 A峰值电流工作模式进行标定）

2.3 运行工况

表1列出了运行工况。

表1. 工况参数

参数	极小点	最大值	单元
工作供电电压	4.4	45	V
操作供电电流（EN_GATE = 低电平）		150	mA
工作温度	–40	125	°C

3 Getting Started

3.1 Requirements

The BOOSTXL-DRV8305EVM is not a standalone motor control kit and requires a compatible XL LaunchPad to provide the appropriate motor control signals. The BOOSTXL-DRV8305EVM has been specifically designed for the LAUNCHXL-F28027F InstaSPIN-FOC LaunchPad. In addition to the BoosterPack and a compatible XL LaunchPad, a 3-phase motor and sufficient power supply are required.

3.2 Configuring the LaunchPad

The BOOSTXL-DRV8305EVM BoosterPack supplies 3.3 V to the LaunchPad through the onboard LMR16006 0.6-A, step-down buck regulator. It is recommended to remove the jumpers on the LaunchPad that connect the emulation and controller power supplies. The LaunchPad communication lines should also be configured to ensure proper operation from the host PC.

Example: LAUNCHXL-F28027F

For the LAUNCHXL-F28027F LaunchPad, remove the JP1 (3.3 V), JP2 (GND), and JP3 (5 V) jumpers to isolate the two power supply domains (MCU and Emulator).

The S1 switch should be set to the ON-ON-ON position to allow for a JTAG debug connection. The S4 switch should be moved to the OFF position to allow for the nFAULT pin from the DRV8305 to report correctly.

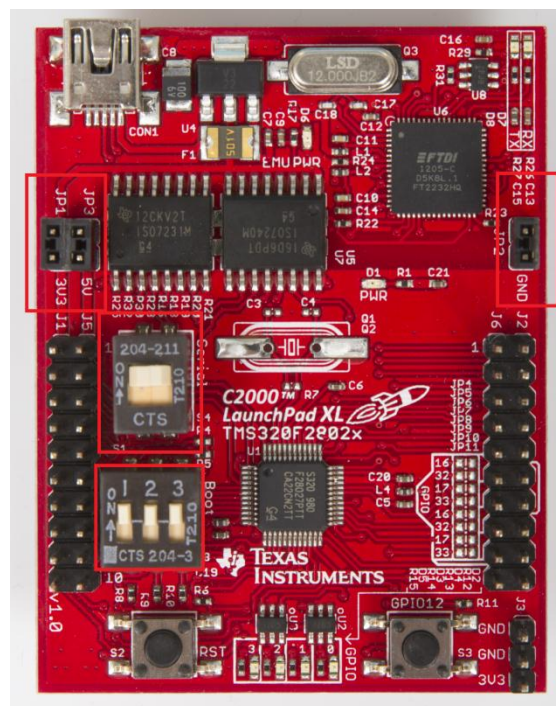


Figure 7. LAUNCHXL-F28027F Configuration

3 快速入门

3.1 需求

boostxl-DRV8305EVM 并非独立的电机控制套件，需搭配兼容的 XL LaunchPad 以提供适当的电机控制信号。该器件专为 launchxl-F28027F InstaSPIN-FOC LaunchPad 设计。除 BoosterPack 和兼容的 XL LaunchPad 外，还需配备三相电机及充足的电源。

3.2 LaunchPad 配置

BoostXL-DRV8305EVM BoosterPack 通过其板载的 LMR16006 0.6 A 降压 Buck 电源调节器为 LaunchPad 提供 3.3 V 电源。建议移除 LaunchPad 上连接仿真电源与控制器电源的跳线。此外，应配置 LaunchPad 的通信线路，以确保其能够与主机 PC 正常通信。

示例：launchxl-F28027F

对于 LaunchPad-F28027F，需拔除 JP1（3.3 V）、JP2（GND）和 JP3（5 V）跳线，以隔离两个电源域（MCU 和 Emulator）。

S1 开关应设置为 ON-ON-ON 位置，以便建立 JTAG 调试连接；S4 开关应切换至 OFF 位置，以确保 DRV 8305 的 nFAULT 引脚能够正确报告故障信息。

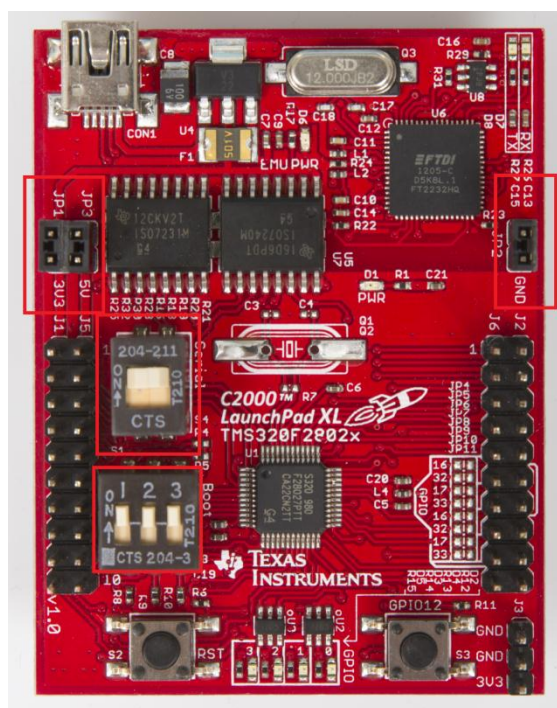


图 7. launchxl-F28027F 配置

3.3 Connecting the Hardware

Use the following steps to connect the hardware:

1. Plug the BOOSTXL-DRV8305EVM BoosterPack onto the LaunchPad as shown in [Figure 5](#). The terminal block headers should be oriented towards the USB connector and the 20-pin headers (J1 and J2) should align properly.
2. Connect the 3-phase motor to the terminal block header J4. The motor connections have been labeled with A, B, and C but can be connected in any order.
3. Connect the power supply, that will power the BoosterPack's DRV8305 3-phase gate driver, 3-phase power stage, and LMR16006 buck regulator to the terminal block header J3. The connections have been labeled PVDD and GND. For full performance, ensure the supply can support as much current as your motor may demand. The BoosterPack has a designed operating range from 4.4- to 45-V with up to 15-A RMS (20-A peak) of phase output current.

WARNING

At high currents the drive stage can increase to high temperatures. Use proper handling procedures.

4. Enable the power supply. A fault may appear on the nFAULT LED. This is normal and should be cleared when the status registers are read or EN_GATE is taken HIGH.
5. Enable the control algorithm and begin spinning the motor. The BOOSTXL-DRV8305EVM BoosterPack combined with a compatible XL LaunchPad will provide a complete motor drive and control evaluation platform. With the Piccolo LAUNCHXL-F28027F LaunchPad you can take full advantage of **TI's InstaSPIN™-FOC sensorless control solution**. To get started with InstaSPIN-FOC (<http://www.ti.com/instaspin-foc>) download and run the MotorWare™ software (<http://www.ti.com/tool/motorware>), reviewing the LAUNCHXL and BOOSTXL resources.

4 Demo Application

The BOOSTXL-DRV8305EVM BoosterPack has been optimized to work together with the Piccolo LAUNCHXL-F28027F LaunchPad to provide a complete motor drive and control evaluation platform. To quickly get your 3-phase motor spinning, see TI's InstaSPIN-FOC sensorless control solution at <http://www.ti.com/tool/motorware>. Multiple projects, labs, and an easy to use GUI are available with TI MotorWare available at <http://www.ti.com/tool/motorware>, with detailed documentation and user guides.

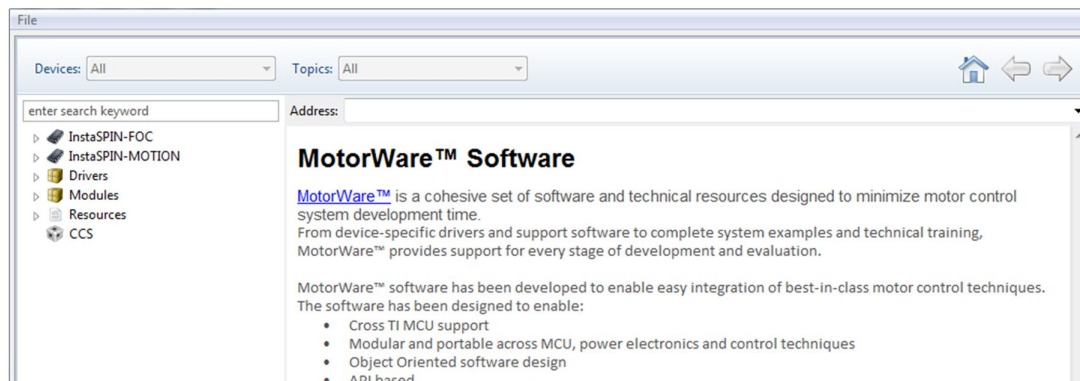


Figure 8. MotorWare™ Software

3.3 连接硬件

请按照以下步骤连接硬件：

1. 如图 5 所示，将 BoostXL-DRV8305EVM BoosterPack 插入 LaunchPad。端子排接线端子应朝向 USB 接口，且 20 针接线端子 (J1 和 J2) 应正确对齐。
2. 将三相电动机连接至接线端子排接线端子 J4。电动机接线端子已标有 A、B 和 C 标记，但可按任意顺序连接。
3. 将电源连接至接线端子块 J3，该电源将为 BoosterPack 中的 DRV8305 三相门极驱动器、三相功率级以及 LMR16006 降压稳压器供电。这些连接端子已分别标为 PVDD 和 GND。为确保系统达到最佳性能，请确认所用电源能够满足电机所需的最大电流需求。BoosterPack 的设计工作电压范围为 4.4–45 V，相输出电流峰值可达 20 A（均方根值为 15 A）。

警告

在大电流工况下，驱动级可能升至高温度。请遵循正确的操作规范。

4. 接通电源。nFAULT LED 上可能会出现故障指示灯。此现象属正常情况，在读取状态寄存器或 EN_GATE 信号变为高电平时应清除该指示灯。
5. 启用控制算法并启动电机。BoostXL-DRV8305EVM BoosterPack 与兼容的 XL LaunchPad 结合使用，可提供一个完整的电机驱动与控制评估平台。通过 Piccolo LaunchXL-F28027F LaunchPad，您可以充分发挥 TI 的 InstaSPIN™-FOC 无传感器控制解决方案的优势。如需开始使用 InstaSPIN-FOC (<http://www.ti.com/instaspin-foc>)，请下载并运行 MotorWare™ 软件。
(<http://www.ti.com/tool/motorware>)，查阅 launchxl 与 boostxl 相关资源。

4 演示应用程序

BoostXL-DRV8305EVM BoosterPack 经过优化，可与 Piccolo LaunchXL-F28027F LaunchPad 协同工作，提供一个完整的电机驱动与控制评估平台。如需快速使您的三相电机启动，请参阅 TI 的 InstaSPIN-FOC 无传感器控制解决方案，网址为 <http://www.ti.com/tool/motorware>。此外，TI MotorWare 提供了多种项目示例、实验教程以及易于使用的图形用户界面 (GUI)，可访问 <http://www.ti.com/tool/motorware> 获取详细文档和用户指南。

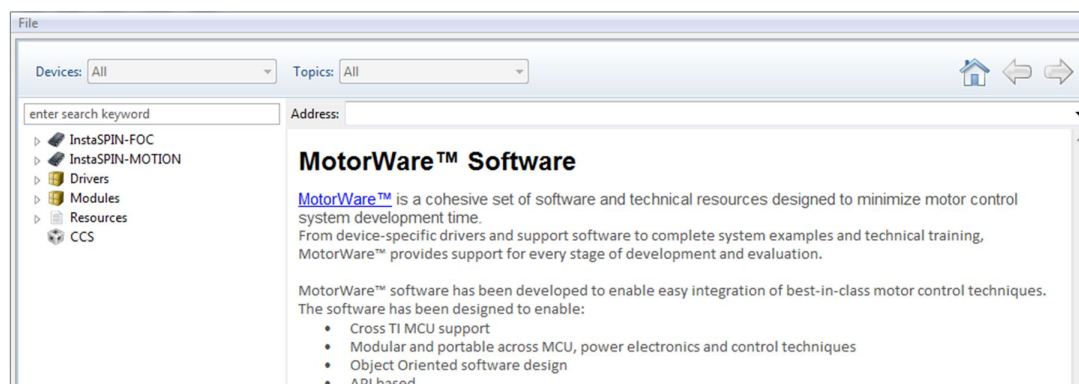


图 8. MotorWare™ 软件

5 Detailed Hardware Description

The BOOSTXL-DRV8305EVM BoosterPack is a complete drive stage for 3-phase motor applications. The design consists of the DRV8305 motor gate driver, CSD18540Q5B N-Channel NexFET Power MOSFETs, and LMR16006 buck regulator. See the respective data sheets for the DRV8305 ([SLVSCX2](#)), CSD18540Q5B ([SLPS488](#)), and LMR16006 ([SNVSA24](#)) for more information concerning each device.

5.1 DC Bus and Phase Voltage Sense

The BoosterPack has been designed with voltage sense circuits on the DC bus (PVDD) and each half-bridge outputs (phases A, B, and C). These circuits, shown [Figure 9](#), consist of a voltage divider with a filtering capacitor to reduce high frequency noise on the ADC pins. These circuits have been scaled to support 4.4 to 45 V. The high-side resistors for the phase outputs are located near the motor output header (J4) while the low-side resistors and filtering capacitors are located near the ADC pins on the BoosterPack to LaunchPad header (J1) for improved noise reduction purposes.

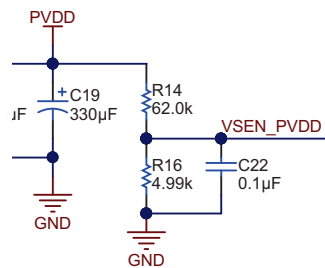


Figure 9. Voltage Sense

To achieve higher resolution voltage feedback, the scaling can be adjusted by replacing the high-side resistors with a lower value.

Example: For a 12-V system, R8, R9, R10, and R14 could be replaced with 22-k Ω resistors to approximately triple the voltage resolution. The new full scale voltage would be 17.85 V.

5.2 Low-Side Current Shunt Sense

The BoosterPack has low-side current shunt sense for each half-bridge (phases A, B, and C). The current sense setup takes advantage of the DRV8305's triple shunt current amplifiers (phases A, B, and c). The configuration for the low-side sense is shown in [Figure 9](#). The differential amplifier senses voltage across a 0.007- Ω power sense resistor with differential connections. The differential voltage is then amplified by 10 V/V and centered at 1.65 V to allow for sensing both positive and negative currents. The sense resistor has been scaled for 0- to 20-A peak currents.

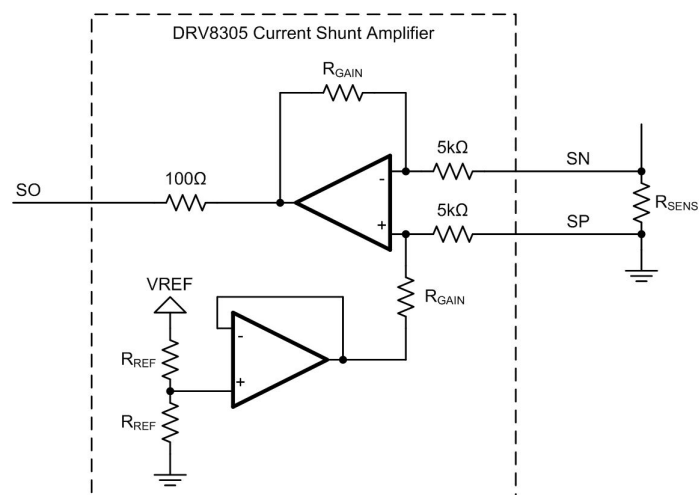


Figure 10. Current Sense

5 详细硬件描述

BoostXL-DRV8305EVM BoosterPack 是一款适用于三相电机应用的完整驱动级解决方案。该设计包含 DRV8305 电机栅极驱动器、CSD18540Q5B N沟道 NexFET 功率 MOSFET 以及 LMR16006 降压稳压器。如需了解有关各器件的更多信息，请参阅相应的数据手册：DRV8305 ([SLVSCX2](#))、CSD18540Q5B ([SLPS488](#)) 及 LMR16006 ([SNVSA24](#))。

5.1 直流母线与相电压检测

BoosterPack 在直流母线 (PVDD) 及每个半桥输出端 (相位 A、B 和 C) 上均配置了电压检测电路。这些电路如图 9 所示，由一个带滤波电容的分压电路组成，用于降低 ADC 引脚上的高频噪声。这些电路经过适配，可支持 4.4 至 45 V 的输入电压范围。相位输出端的高侧电阻器布置在电机输出接线端子 (J4) 附近；而低侧电阻器与滤波电容则布置在 BoosterPack 与 LaunchPad 接线端子 (J1) 之间的 ADC 引脚附近，以实现更优的噪声抑制效果。

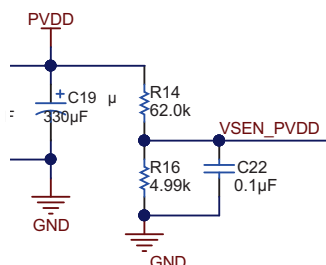


图 9. 电压检测

为实现更高分辨率的电压反馈，可通过将高侧电阻替换为较小阻值来调整缩放系数。

示例：对于 12 V 电源系统，可将 R8、R9、R10 和 R14 替换为 22 kΩ 的电阻器，从而将电压分辨率提高约三倍。新的满量程电压为 17.85 V。

5.2 低侧电流分流检测

BoosterPack 为每个半桥 (相位 A、B 和 C) 配置了低侧电流分流检测电路。该电流检测方案利用了 DRV8305 的三重分流电流放大器 (对应相位 A、B 和 C)。低侧检测电路的配置如图 9 所示。差分放大器通过一个阻值为 0.007 Ω、采用差分连接方式的功率检测电阻来检测其两端电压。该差分电压随后经 10 V/V 放大，并偏置至 1.65 V，以便同时检测正向和负向电流。该检测电阻已针对 0 至 20 A 的峰值电流范围进行了标定。

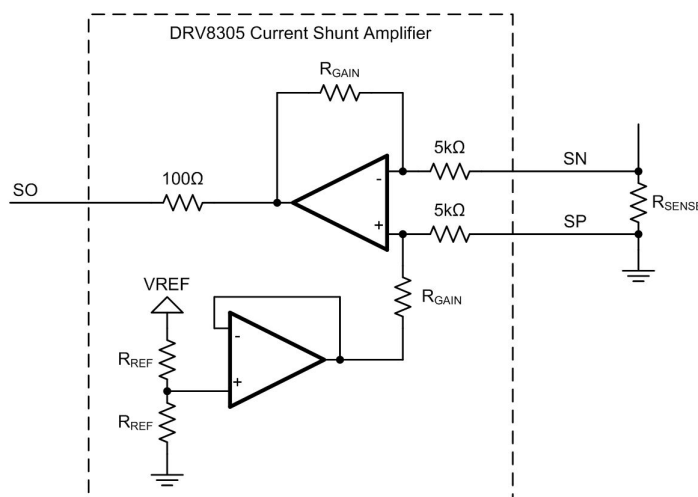


图 10. 电流检测

5.3 BoosterPack GPIO Signals

The Motor Drive BoosterPack brings out the GPIO signals for the DRV8301 to the LaunchPad XL. These signals are described in detail in the following list and further information can be found in the DRV8301 data sheet.

1. **nFAULT:** Fault indicator, specific FAULT status can be obtained through the status registers
2. **PWRGD:** Watchdog and LDO regulator status indicator
3. **EN_GATE:** Enables gate driver and current shunt amplifiers
4. **WAKE:** Used to bring the device out of its low power sleep mode.

5.4 DRV8305 Status and Control Registers

The DRV8305 provides extensive fault reporting and device configuration through an SPI interface and internal registers. There are two categories of registers: status and control. Status registers provide information about device faults and warnings. This information can include items ranging from IC overtemperature to MOSFET overcurrent events. The control registers allow various device parameters to be modified to suit system requirements. These parameters include but are not limited to gate drive current, dead times, current shunt amplifier configurations, and fault reporting modes. For specific information concerning the DRV8305 registers, refer to the data sheet ([SLVSCX2](#)).

The InstaSPIN GUI allows easy access to read and modify the DRV8305's internal registers. These can be accessed on the DRV8305 tab of the InstaSPIN Universal GUI.

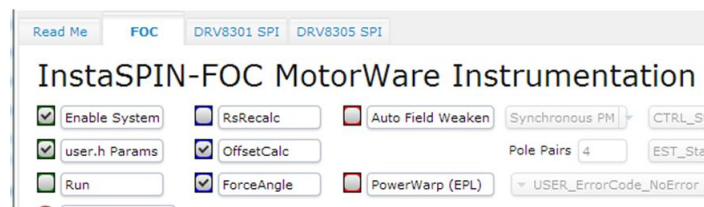


Figure 11. Enable System

5.3 BoosterPack GPIO 信号

Motor Drive BoosterPack 将 DRV8301 的 GPIO 信号输出至 LaunchPad XL。这些信号在下表中进行了详细说明；如需更多信息，请参阅 DRV8301 数据手册。

1. **故障**：故障指示器；可通过状态寄存器获取具体的故障状态
2. **PWRGD**：Watchdog 与 LDO 监管状态指示器
3. **EN_GATE**：启用门驱动器与电流分流放大器
4. **唤醒**：用于将设备从低功耗睡眠模式中唤醒。

5.4 DRV8305 状态与控制寄存器

DRV8305 通过 SPI 接口和内部寄存器提供全面的故障报告与设备配置功能。寄存器分为两类：状态寄存器与控制寄存器。状态寄存器用于提供有关设备故障及警告的信息；这些信息涵盖范围广泛，从 IC 过温事件到 MOSFET 过流事件等。控制寄存器允许对各项设备参数进行修改，以适应系统需求；这些参数包括但不限于栅极驱动电流、死区时间、电流分流放大器配置以及故障报告模式。如需获取有关 DRV8305 寄存器的详细信息，请参阅数据手册 ([SLVSCX2](#))。

InstaSPIN 图形用户界面 (GUI) 可便捷地读取和修改 DRV8305 的内部寄存器；这些寄存器可通过 InstaSPIN 通用图形用户界面中的“DRV8305”选项卡进行访问。

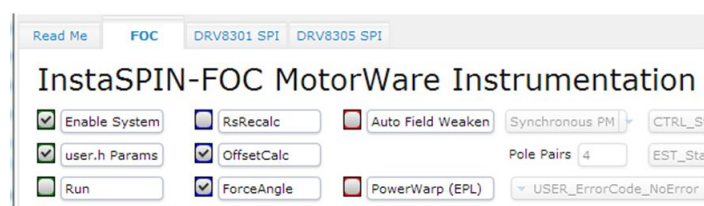
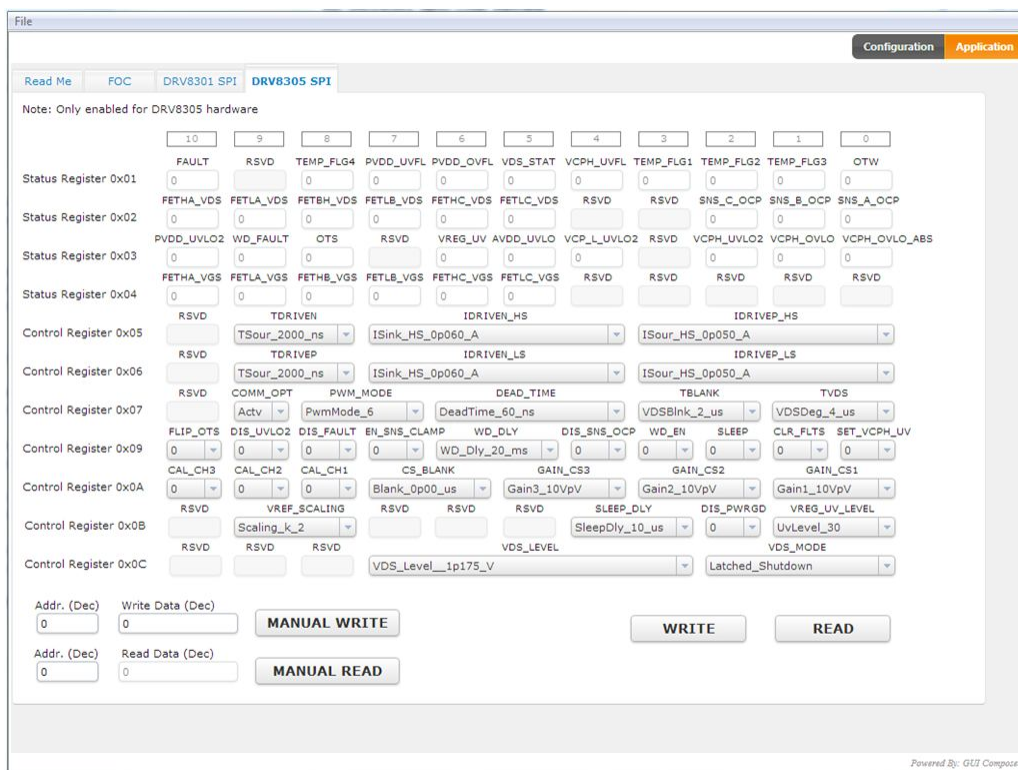


图 11. 系统启用功能

Power to the BoosterPack needs to be supplied and **Enable System** needs to be checked to allow SPI reads and writes. The DRV8305 SPI tab displays a map of the DRV8305 internal registers, both status and control. To read from the registers, select the **Read** button. The register map will update with the current register values. To write to the registers, make the desired change in the map and select the **Write** button. Manual reads or writes to individual registers can be made with the **Manual Write** and **Manual Read** buttons (in decimal).



File Configuration Application

Read Me FOC DRV8301 SPI **DRV8305 SPI**

Note: Only enabled for DRV8305 hardware

Register	10	9	8	7	6	5	4	3	2	1	0
Status Register 0x01	FAULT	RSVD	TEMP_FLG4	PVDD_UVFL	PVDD_OVFL	VDS_STAT	VCPH_UVFL	TEMP_FLG1	TEMP_FLG2	TEMP_FLG3	OTW
Status Register 0x02	FETHA_VDS	FETLA_VDS	FETBH_VDS	FETLB_VDS	FETHC_VDS	FETLC_VDS	RSVD	RSVD	SNS_C_OCP	SNS_B_OCP	SNS_A_OCP
Status Register 0x03	PVDD_UVLO2	WD_FAULT	OTS	RSVD	VREG_UV	AVDD_UVLO	VCP_L_UVLO2	RSVD	VCPH_UVLO2	VCPH_OVLO	VCPH_OVLO_ABS
Status Register 0x04	FETHA_VGS	FETLA_VGS	FETBH_VGS	FETLB_VGS	FETHC_VGS	FETLC_VGS	RSVD	RSVD	RSVD	RSVD	RSVD
Control Register 0x05	RSVD	TORIVEN	IDRIVEN_HS	IDRIVEN_HS	IDRIVEN_HS	IDRIVEN_HS	IDRIVEN_HS	IDRIVEN_HS	IDRIVEN_HS	IDRIVEN_HS	IDRIVEN_HS
Control Register 0x06	RSVD	TORIVEP	IDRIVEN_LS	IDRIVEN_LS	IDRIVEN_LS	IDRIVEN_LS	IDRIVEN_LS	IDRIVEN_LS	IDRIVEN_LS	IDRIVEN_LS	IDRIVEN_LS
Control Register 0x07	RSVD	COMM_OPT	PWM_MODE	DEAD_TIME	DEAD_TIME	DEAD_TIME	DEAD_TIME	DEAD_TIME	DEAD_TIME	DEAD_TIME	DEAD_TIME
Control Register 0x09	FLIP_OTS	DIS_UVLO2	DIS_FAULT	EN_SNS_CLAMP	WD_DLY	DIS_SNS_OCP	WD_EN	SLEEP	CLR_FTS	SET_VCPH_UV	SET_VCPH_UV
Control Register 0x0A	CAL_CH3	CAL_CH2	CAL_CH1	CS_BLANK	GAIN_CS3	GAIN_CS2	GAIN_CS1	GAIN_CS1	GAIN_CS1	GAIN_CS1	GAIN_CS1
Control Register 0x0B	RSVD	VREF_SCALING	RSVD	RSVD	RSVD	SLEEP_DLY	DIS_PWRGD	VREG_UV_LEVEL	VREG_UV_LEVEL	VREG_UV_LEVEL	VREG_UV_LEVEL
Control Register 0x0C	RSVD	RSVD	RSVD	VDS_LEVEL	VDS_LEVEL	SleepDly_10_us	0	UvLevel_30	UvLevel_30	UvLevel_30	UvLevel_30

Addr. (Dec) Write Data (Dec) 0 0 **MANUAL WRITE** **WRITE** **READ**

Addr. (Dec) Read Data (Dec) 0 0 **MANUAL READ**

Powered By: GUI Composer

Figure 12. DRV8305 SPI Registers

需为 BoosterPack 提供电源，并检查“系统启用”选项以允许进行 SPI 读写操作。DRV8305 的 SPI 选项卡显示了 DRV8305 内部寄存器（包括状态寄存器和控制寄存器）的映射表。如需从寄存器读取数据，请点击“读取”按钮；此时寄存器映射表将更新为当前寄存器值。如需向寄存器写入数据，请在映射表中进行所需设置后点击“写入”按钮。如需对单个寄存器进行手动读写操作，可使用“手动写入”和“手动读取”按钮（以十进制形式操作）。

The screenshot displays the DRV8305 SPI configuration tool. At the top, there are tabs for 'Read Me', 'FOC', 'DRV8301 SPI', and 'DRV8305 SPI'. Below the tabs, a note states: 'Note: Only enabled for DRV8305 hardware'. The main area contains a table of registers and their bit fields. The registers are: Status Register 0x01, Status Register 0x02, Status Register 0x03, Status Register 0x04, Control Register 0x05, Control Register 0x06, Control Register 0x07, Control Register 0x09, Control Register 0x0A, Control Register 0x0B, and Control Register 0x0C. Each register has a set of bit fields with corresponding values. For example, Status Register 0x01 has bit fields: 10 (FAULT), 9 (RSVD), 8 (TEMP_FLG4), 7 (PVDD_UVFL), 6 (PVDD_OVFL), 5 (VDS_STAT), 4 (VCPH_UVFL), 3 (TEMP_FLG1), 2 (TEMP_FLG2), 1 (TEMP_FLG3), and 0 (OTW). Below the table, there are buttons for 'MANUAL WRITE', 'WRITE', 'READ', 'MANUAL READ', and 'WRITE DATA'. The interface is titled 'DRV8305 SPI' and includes a 'Configuration' tab.

图 12. DRV8305 SPI 寄存器

5.5 BOOSTXL-DRV8305EVM Schematic

Figure 13 shows the BOOSTXL-DRV8305EVM schematic.

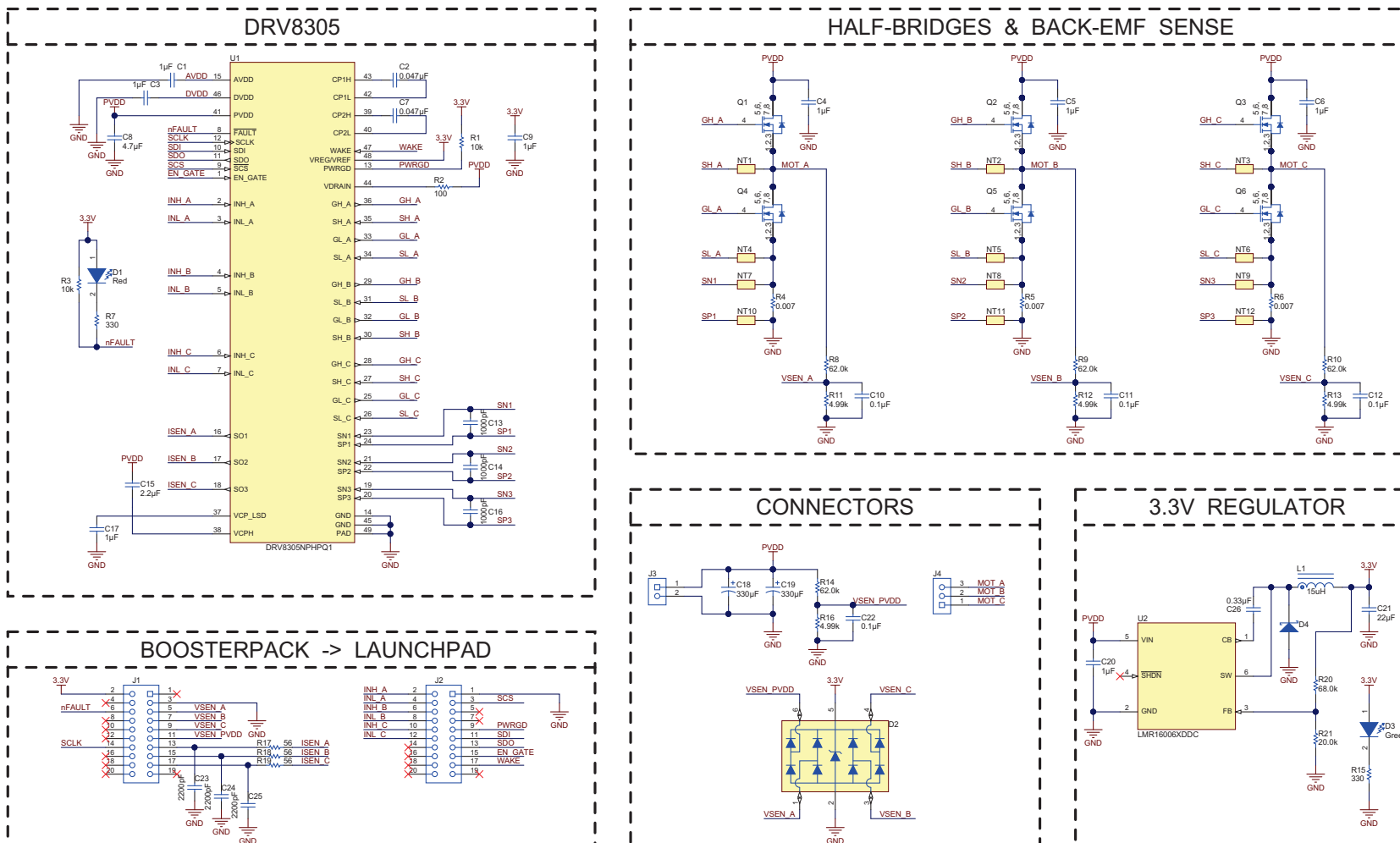


Figure 13. BOOSTXL-DRV8305EVM Schematic

5.5 boostxl-DRV8305EVM原理图

图 13展示了 boostxl-DRV8305EVM 的原理图。

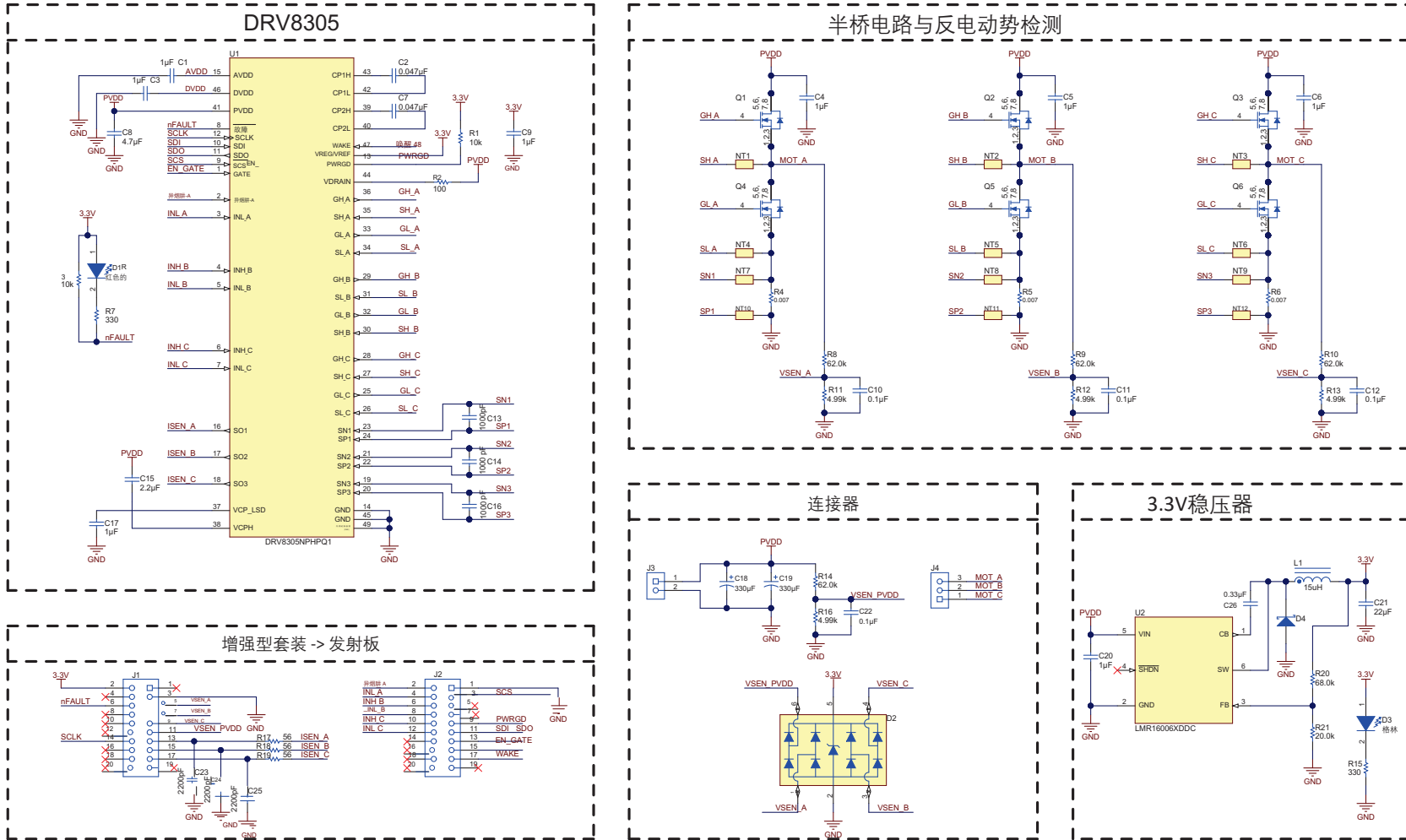


图 13. boostxl.DRV8305EVM 方框图

5.6 Hardware Source Files

The complete design files can be found on the tool folder, including the schematic, Gerbers, designs files, PCB views, and bill of materials.

5.6 硬件源文件

完整的设计文件可从“工具”文件夹中获取，包括原理图、Gerber文件、设计文件、PCB视图及物料清单。

Revision History

NOTE: Page numbers for previous revisions may differ from page numbers in the current version.

Changes from Original (August 2015) to A Revision	Page
---	------

- | | |
|---|---|
| • Added the <i>Operating Conditions</i> section | 5 |
|---|---|

版本历史

注：此前版本中的页码可能与当前版本中的页码有所不同。

从原版（2015年8月）至修订版的变更内容	页
• 新增了“运行工况”部分	5

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 - 3.1.1 **Notice applicable to EVMs not FCC-Approved:**

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 - 3.1.2 **For EVMs annotated as FCC – FEDERAL COMMUNICATIONS COMMISSION Part 15 Compliant:**

CAUTION

This device complies with part 15 of the FCC Rules. Operation is subject to the following two conditions: (1) This device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation.

Changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment.

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- 3 监管公告:
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谨慎

本设备符合《FCC 规则》第 15 部分的规定。其操作须满足以下两项条件: (1) 本设备不得产生有害干扰; (2) 本设备必须能够接受所接收的任何干扰, 包括可能导致非预期操作的干扰。

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A 类 EVM 设备的面心立方干扰声明

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FCC Interference Statement for Class B EVM devices

NOTE: This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference in a residential installation. This equipment generates, uses and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation. If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures:

- *Reorient or relocate the receiving antenna.*
- *Increase the separation between the equipment and receiver.*
- *Connect the equipment into an outlet on a circuit different from that to which the receiver is connected.*
- *Consult the dealer or an experienced radio/TV technician for help.*

3.2 Canada

3.2.1 For EVMs issued with an Industry Canada Certificate of Conformance to RSS-210 or RSS-247

Concerning EVMs Including Radio Transmitters:

This device complies with Industry Canada license-exempt RSSs. Operation is subject to the following two conditions:

(1) this device may not cause interference, and (2) this device must accept any interference, including interference that may cause undesired operation of the device.

Concernant les EVMs avec appareils radio:

Le présent appareil est conforme aux CNR d'Industrie Canada applicables aux appareils radio exempts de licence. L'exploitation est autorisée aux deux conditions suivantes: (1) l'appareil ne doit pas produire de brouillage, et (2) l'utilisateur de l'appareil doit accepter tout brouillage radioélectrique subi, même si le brouillage est susceptible d'en compromettre le fonctionnement.

Concerning EVMs Including Detachable Antennas:

Under Industry Canada regulations, this radio transmitter may only operate using an antenna of a type and maximum (or lesser) gain approved for the transmitter by Industry Canada. To reduce potential radio interference to other users, the antenna type and its gain should be so chosen that the equivalent isotropically radiated power (e.i.r.p.) is not more than that necessary for successful communication. This radio transmitter has been approved by Industry Canada to operate with the antenna types listed in the user guide with the maximum permissible gain and required antenna impedance for each antenna type indicated. Antenna types not included in this list, having a gain greater than the maximum gain indicated for that type, are strictly prohibited for use with this device.

Concernant les EVMs avec antennes détachables

Conformément à la réglementation d'Industrie Canada, le présent émetteur radio peut fonctionner avec une antenne d'un type et d'un gain maximal (ou inférieur) approuvé pour l'émetteur par Industrie Canada. Dans le but de réduire les risques de brouillage radioélectrique à l'intention des autres utilisateurs, il faut choisir le type d'antenne et son gain de sorte que la puissance isotrope rayonnée équivalente (p.i.r.e.) ne dépasse pas l'intensité nécessaire à l'établissement d'une communication satisfaisante. Le présent émetteur radio a été approuvé par Industrie Canada pour fonctionner avec les types d'antenne énumérés dans le manuel d'usage et ayant un gain admissible maximal et l'impédance requise pour chaque type d'antenne. Les types d'antenne non inclus dans cette liste, ou dont le gain est supérieur au gain maximal indiqué, sont strictement interdits pour l'exploitation de l'émetteur.

3.3 Japan

3.3.1 *Notice for EVMs delivered in Japan:* Please see http://www.tij.co.jp/llds/ti_ja/general/eStore/notice_01.page 日本国内に輸入される評価用キット、ボードについては、次のところをご覧ください。
http://www.tij.co.jp/llds/ti_ja/general/eStore/notice_01.page

3.3.2 *Notice for Users of EVMs Considered "Radio Frequency Products" in Japan:* EVMs entering Japan may not be certified by TI as conforming to Technical Regulations of Radio Law of Japan.

If User uses EVMs in Japan, not certified to Technical Regulations of Radio Law of Japan, User is required to follow the instructions set forth by Radio Law of Japan, which includes, but is not limited to, the instructions below with respect to EVMs (which for the avoidance of doubt are stated strictly for convenience and should be verified by User):

1. Use EVMs in a shielded room or any other test facility as defined in the notification #173 issued by Ministry of Internal Affairs and Communications on March 28, 2006, based on Sub-section 1.1 of Article 6 of the Ministry's Rule for Enforcement of Radio Law of Japan,
2. Use EVMs only after User obtains the license of Test Radio Station as provided in Radio Law of Japan with respect to EVMs, or
3. Use of EVMs only after User obtains the Technical Regulations Conformity Certification as provided in Radio Law of Japan with respect to EVMs. Also, do not transfer EVMs, unless User gives the same notice above to the transferee. Please note that if User does not follow the instructions above, User will be subject to penalties of Radio Law of Japan.

B类 EVM 设备的面心立方干扰声明

注意：本设备经测试，符合《FCC规则》第15部分对B类数字设备所规定的限值要求。这些限值旨在为住宅环境中的设备安装提供合理的抗有害干扰保护。本设备会产生、使用并可能辐射射频能量；若未按照说明书要求进行安装和使用，可能对无线电通信造成有害干扰。然而，无法保证在特定安装环境中不会发生干扰。若本设备确实对无线电或电视接收造成有害干扰（可通过关闭后重新开启设备来判断），建议用户尝试采取以下一项或多项措施来消除该干扰：

- 重新定向或移动接收天线。
- 增加设备与接收机之间的间距。
- 将设备连接到与接收机所连接的电路不同的电源插座上。
- 请咨询经销商或经验丰富的无线电/电视技术员以获取帮助。

3.2 加拿大

3.2.1 适用于持有加拿大工业部颁发的符合 RSS-210 或 RSS-247 标准之合格证书的电子验证模块 (EVM)

关于包含无线电发射机的电子万用表 (EVM)：

本设备符合加拿大工业部免许可 RSS 标准。其操作须满足以下两项条件：

(1) 该设备不得产生干扰；且(2)该设备必须能够承受任何干扰，包括可能导致设备出现非预期操作的干扰。

关于配备无线电设备的 EVM：

本设备符合加拿大工业部适用于免许可无线电设备的适用性法规。在满足以下两项条件时，允许对该设备使用：(1) 该设备不得产生无线电干扰；(2) 设备用户应接受设备所遭受的任何无线电干扰，即使该干扰可能影响设备的正常运行。

关于包含可拆卸天线的电子万用表：

根据加拿大工业部的相关法规，本无线电发射机仅可在使用经加拿大工业部批准的、具有特定类型及最大（或较低）增益的天线时进行操作。为减少对其他用户的潜在无线电干扰，应选择合适天线类型及其增益，确保其等效各向同性辐射功率（EIRP）不超过实现有效通信所需的最小值。本无线电发射机已获得加拿大工业部的批准，可配合用户手册中列出的天线类型使用；每种天线类型均标明了其允许的最大增益及所需天线阻抗。对于未列入本清单、且其增益超过该类型所标最大增益的天线类型，严禁用于本设备。

适用于配备可拆卸天线的EVM设备

根据加拿大工业部的法规要求，本无线电发射机可与经加拿大工业部批准、具有特定类型及最大（或较低）增益的天线配合使用。为降低对其他用户造成的无线电干扰风险，用户应选择合适天线类型及增益，确保其等效各向同性辐射功率（p.i.r.e.）不超过建立满意通信所需的强度。本无线电发射机经加拿大工业部批准，可与用户手册中列出的、具有允许最大增益及对应于各天线类型的所需阻抗的天线类型配合使用。对于未列入本清单或其增益超过允许最大值的天线类型，严禁用于本发射机的运行。

3.3 日本

3.3.1 针对运抵日本的EVMs的通知：请参阅http://www.tij.co.jp/lstds/ti_ja/general/eStore/notice_01.page《关于进口至日本的评估用键盘及鼠标产品之说明》，详见下文。
http://www.tij.co.jp/lstds/ti_ja/general/eStore/notice_01.page

3.3.2 日本境内将EVM视为“射频产品”的用户须知：进入日本市场的EVM产品可能未获得TI对其符合《日本无线电法》技术法规的认证。

若用户在日本境内使用未获得《日本无线电法》技术法规认证的EVM设备，则用户须遵守《日本无线电法》所规定的各项指令；这些指令包括但不限于下文针对EVM设备所列的指令（为免疑义，上述指令仅为方便起见而列示，用户应自行核实）：

1. 在屏蔽室或任何其他符合内务省与通信省于2006年3月28日发布的第173号通知所规定的测试设施中使用EVM；该通知系依据内务省《日本无线电法执行规则》第6条第1.1款制定。
2. 用户仅可在取得根据《日本无线电法》关于EVM的规定颁发的测试无线电台许可证后，方可使用EVM；或
3. 用户须在取得根据《日本无线电法》对EVMs所规定的《技术法规符合性证书》后，方可使用EVMs；此外，未经用户向受让方发出上述相同通知，不得转让EVMs。请注意，若用户未遵守上述规定，用户将依照《日本无线电法》受到处罚。

【無線電波を送信する製品の開発キットをお使いになる際の注意事項】 開発キットの中には技術基準適合証明を受けていないものがあります。技術適合証明を受けていないもののご使用に際しては、電波法遵守のため、以下のいずれかの措置を取っていただく必要がありますのでご注意ください。

1. 電波法施行規則第6条第1項第1号に基づく平成18年3月28日総務省告示第173号で定められた電波暗室等の試験設備でご使用いただく。
2. 実験局の免許を取得後ご使用いただく。
3. 技術基準適合証明を取得後ご使用いただく。

なお、本製品は、上記の「ご使用にあたっての注意」を譲渡先、移転先に通知しない限り、譲渡、移転できないものとします。

上記を遵守頂けない場合は、電波法の罰則が適用される可能性があることをご留意ください。日本テキサス・インスツルメンツ株式会社
東京都新宿区西新宿 6 丁目 2 4 番 1 号
西新宿三井ビル

3.3.3 *Notice for EVMs for Power Line Communication:* Please see http://www.tij.co.jp/lstds/ti_ja/general/eStore/notice_02.page
電力線搬送波通信についての開発キットをお使いになる際の注意事項については、次のところをご覧ください。 http://www.tij.co.jp/lstds/ti_ja/general/eStore/notice_02.page

3.4 *European Union*

3.4.1 *For EVMs subject to EU Directive 2014/30/EU (Electromagnetic Compatibility Directive):*

This is a class A product intended for use in environments other than domestic environments that are connected to a low-voltage power-supply network that supplies buildings used for domestic purposes. In a domestic environment this product may cause radio interference in which case the user may be required to take adequate measures.

4 *EVM Use Restrictions and Warnings:*

4.1 EVMS ARE NOT FOR USE IN FUNCTIONAL SAFETY AND/OR SAFETY CRITICAL EVALUATIONS, INCLUDING BUT NOT LIMITED TO EVALUATIONS OF LIFE SUPPORT APPLICATIONS.

4.2 User must read and apply the user guide and other available documentation provided by TI regarding the EVM prior to handling or using the EVM, including without limitation any warning or restriction notices. The notices contain important safety information related to, for example, temperatures and voltages.

4.3 *Safety-Related Warnings and Restrictions:*

4.3.1 User shall operate the EVM within TI's recommended specifications and environmental considerations stated in the user guide, other available documentation provided by TI, and any other applicable requirements and employ reasonable and customary safeguards. Exceeding the specified performance ratings and specifications (including but not limited to input and output voltage, current, power, and environmental ranges) for the EVM may cause personal injury or death, or property damage. If there are questions concerning performance ratings and specifications, User should contact a TI field representative prior to connecting interface electronics including input power and intended loads. Any loads applied outside of the specified output range may also result in unintended and/or inaccurate operation and/or possible permanent damage to the EVM and/or interface electronics. Please consult the EVM user guide prior to connecting any load to the EVM output. If there is uncertainty as to the load specification, please contact a TI field representative. During normal operation, even with the inputs and outputs kept within the specified allowable ranges, some circuit components may have elevated case temperatures. These components include but are not limited to linear regulators, switching transistors, pass transistors, current sense resistors, and heat sinks, which can be identified using the information in the associated documentation. When working with the EVM, please be aware that the EVM may become very warm.

4.3.2 EVMs are intended solely for use by technically qualified, professional electronics experts who are familiar with the dangers and application risks associated with handling electrical mechanical components, systems, and subsystems. User assumes all responsibility and liability for proper and safe handling and use of the EVM by User or its employees, affiliates, contractors or designees. User assumes all responsibility and liability to ensure that any interfaces (electronic and/or mechanical) between the EVM and any human body are designed with suitable isolation and means to safely limit accessible leakage currents to minimize the risk of electrical shock hazard. User assumes all responsibility and liability for any improper or unsafe handling or use of the EVM by User or its employees, affiliates, contractors or designees.

4.4 User assumes all responsibility and liability to determine whether the EVM is subject to any applicable international, federal, state, or local laws and regulations related to User's handling and use of the EVM and, if applicable, User assumes all responsibility and liability for compliance in all respects with such laws and regulations. User assumes all responsibility and liability for proper disposal and recycling of the EVM consistent with all applicable international, federal, state, and local requirements.

5. *Accuracy of Information:* To the extent TI provides information on the availability and function of EVMs, TI attempts to be as accurate as possible. However, TI does not warrant the accuracy of EVM descriptions, EVM availability or other information on its websites as accurate, complete, reliable, current, or error-free.

【無線電波を送信する製品の開発キットをお使いになる際の注意事項】 開発キットの中には技術基準適合証明を受けていないものがあります。技術適合証明を受けていないもののご使用に際しては、電波法遵守のため、以下のいずれかの措置を取っていただく必要がありますのでご注意ください。

1. 電波法施行規則第6条第1項第1号に基づく平成18年3月28日総務省告示第173号で定められた電波暗室等の試験設備でご使用いただく。
2. 実験局の免許を取得後ご使用いただく。
3. 技術基準適合証明を取得後ご使用いただく。

なお、本製品は、上記の「ご使用にあたっての注意」を譲渡先、移転先に通知しない限り、譲渡、移転できないものとします。

上記を遵守頂けない場合は、電波法の罰則が適用される可能性があることをご留意ください。日本テキサス・インスツルメンツ株式会社
東京都新宿区西新宿 6 丁目 2 4 番 1 号
西新宿三井ビル

3.3.3 电力线通信用EVM用户须知: 如需了解http://www.tij.co.jp/lsds/ti_ja/general/eStore/notice_02.page “电力线传输波通信开发要点” 中的注意事项, 请参阅以下页面: http://www.tij.co.jp/lsds/ti_ja/general/eStore/notice_02.page

3.4 欧洲联盟

3.4.1 适用于受欧盟第2014/30/EU号指令(电磁兼容性指令)约束的电子设备:

本产品属于A类产品, 适用于除住宅环境之外的其他环境; 此类环境需连接至为住宅用途建筑供电的低压电源网络。在住宅环境中, 本产品可能会引发无线电干扰, 此时用户可能需采取适当的防护措施。

4 EVM 使用限制与警告:

4.1 EVMS 不适用于功能安全及/或安全关键性评估 (包括但不限于对生命支持应用的评估)。

4.2 用户在操作或使用 EVM 之前, 必须阅读并遵守 TI 提供的关于 EVM 的用户手册及其他相关文档 (包括但不限于任何警告或限制性通知)。这些通知包含与温度、电压等相关的重要安全信息。

4.3 安全相关警告与限制:

4.3.1 用户应按照 TI 在用户手册、TI 提供的其他可用文档以及任何其他适用要求中所规定的推荐规格和环境考量要求, 对 EVM 进行操作, 并采取合理且符合惯例的防护措施。若超出 EVM 的指定性能等级和规格 (包括但不限于输入/输出电压、电流、功率及环境参数范围), 可能导致人员伤亡或财产损失。如对性能等级及规格存在疑问, 用户应在连接包括输入电源及预期负载在内的接口电子设备之前, 联系 TI 的现场代表。任何施加在超出指定输出范围内的负载都可能导致非预期或不准确的运行, 或对 EVM 及/或接口电子设备造成永久性损坏。在将任何负载连接至 EVM 输出端之前, 请参阅 EVM 用户手册。若对负载规格存在疑问, 请联系 TI 的现场代表。在正常运行过程中, 即使输入和输出信号均保持在规定的允许范围内, 部分电路元件的外壳温度仍可能偏高。这些元件包括但不限于线性稳压器、开关晶体管、导通晶体管、电流检测电阻器及散热片等; 可通过相关技术文档中的信息识别这些元件。在操作 EVM 时, 请注意该设备可能会变得非常烫。

4.3.2 EVM 仅限具备相应技术资质的专业电子技术专家使用; 此类专家应熟悉操作机电组件、系统及子系统所涉及的危险性及应用风险。用户或其员工、关联公司、承包商或指定人员在对 EVM 进行正确且安全的操作与使用过程中, 应承担全部责任与法律责任。用户有责任确保 EVM 与人体之间所有的接口 (电子及/或机械接口) 均采用适当的绝缘措施及有效手段对可触及的漏电流进行安全限制, 从而最大限度降低触电危险。对于用户或其员工、关联公司、承包商或指定人员对 EVM 进行任何不当或不安全的操作或使用行为, 用户应承担全部责任与法律责任。

4.4 用户应自行承担全部责任与义务, 以确定该 EVM 是否受任何适用于用户对 EVM 进行处理和使用的国际、联邦、州或地方法律及法规的约束; 如适用, 用户应承担确保其在所有方面均符合上述法律及法规要求的全部责任与义务。用户还应承担对 EVM 进行妥善处置和回收利用的全部责任与义务, 且该处置与回收利用行为须符合所有适用的国际、联邦、州及地方法律要求。

5. 信息准确性: 在 TI 提供有关 EVMS 的可用性及功能信息时, TI 将力求确保这些信息尽可能准确。然而, TI 不保证其网站上所载的 EVM 描述、EVM 可用性或其他信息的准确性、完整性、可靠性、时效性或无错误性。

6. *Disclaimers:*

6.1 EXCEPT AS SET FORTH ABOVE, EVMS AND ANY MATERIALS PROVIDED WITH THE EVM (INCLUDING, BUT NOT LIMITED TO, REFERENCE DESIGNS AND THE DESIGN OF THE EVM ITSELF) ARE PROVIDED "AS IS" AND "WITH ALL FAULTS." TI DISCLAIMS ALL OTHER WARRANTIES, EXPRESS OR IMPLIED, REGARDING SUCH ITEMS, INCLUDING BUT NOT LIMITED TO ANY EPIDEMIC FAILURE WARRANTY OR IMPLIED WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE OR NON-INFRINGEMENT OF ANY THIRD PARTY PATENTS, COPYRIGHTS, TRADE SECRETS OR OTHER INTELLECTUAL PROPERTY RIGHTS.

6.2 EXCEPT FOR THE LIMITED RIGHT TO USE THE EVM SET FORTH HEREIN, NOTHING IN THESE TERMS SHALL BE CONSTRUED AS GRANTING OR CONFERRING ANY RIGHTS BY LICENSE, PATENT, OR ANY OTHER INDUSTRIAL OR INTELLECTUAL PROPERTY RIGHT OF TI, ITS SUPPLIERS/LICENSORS OR ANY OTHER THIRD PARTY, TO USE THE EVM IN ANY FINISHED END-USER OR READY-TO-USE FINAL PRODUCT, OR FOR ANY INVENTION, DISCOVERY OR IMPROVEMENT, REGARDLESS OF WHEN MADE, CONCEIVED OR ACQUIRED.

7. *USER'S INDEMNITY OBLIGATIONS AND REPRESENTATIONS.* USER WILL DEFEND, INDEMNIFY AND HOLD TI, ITS LICENSORS AND THEIR REPRESENTATIVES HARMLESS FROM AND AGAINST ANY AND ALL CLAIMS, DAMAGES, LOSSES, EXPENSES, COSTS AND LIABILITIES (COLLECTIVELY, "CLAIMS") ARISING OUT OF OR IN CONNECTION WITH ANY HANDLING OR USE OF THE EVM THAT IS NOT IN ACCORDANCE WITH THESE TERMS. THIS OBLIGATION SHALL APPLY WHETHER CLAIMS ARISE UNDER STATUTE, REGULATION, OR THE LAW OF TORT, CONTRACT OR ANY OTHER LEGAL THEORY, AND EVEN IF THE EVM FAILS TO PERFORM AS DESCRIBED OR EXPECTED.

8. *Limitations on Damages and Liability:*

8.1 *General Limitations.* IN NO EVENT SHALL TI BE LIABLE FOR ANY SPECIAL, COLLATERAL, INDIRECT, PUNITIVE, INCIDENTAL, CONSEQUENTIAL, OR EXEMPLARY DAMAGES IN CONNECTION WITH OR ARISING OUT OF THESE TERMS OR THE USE OF THE EVMS, REGARDLESS OF WHETHER TI HAS BEEN ADVISED OF THE POSSIBILITY OF SUCH DAMAGES. EXCLUDED DAMAGES INCLUDE, BUT ARE NOT LIMITED TO, COST OF REMOVAL OR REINSTALLATION, ANCILLARY COSTS TO THE PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES, RETESTING, OUTSIDE COMPUTER TIME, LABOR COSTS, LOSS OF GOODWILL, LOSS OF PROFITS, LOSS OF SAVINGS, LOSS OF USE, LOSS OF DATA, OR BUSINESS INTERRUPTION. NO CLAIM, SUIT OR ACTION SHALL BE BROUGHT AGAINST TI MORE THAN TWELVE (12) MONTHS AFTER THE EVENT THAT GAVE RISE TO THE CAUSE OF ACTION HAS OCCURRED.

8.2 *Specific Limitations.* IN NO EVENT SHALL TI'S AGGREGATE LIABILITY FROM ANY USE OF AN EVM PROVIDED HEREUNDER, INCLUDING FROM ANY WARRANTY, INDEMNITY OR OTHER OBLIGATION ARISING OUT OF OR IN CONNECTION WITH THESE TERMS, EXCEED THE TOTAL AMOUNT PAID TO TI BY USER FOR THE PARTICULAR EVM(S) AT ISSUE DURING THE PRIOR TWELVE (12) MONTHS WITH RESPECT TO WHICH LOSSES OR DAMAGES ARE CLAIMED. THE EXISTENCE OF MORE THAN ONE CLAIM SHALL NOT ENLARGE OR EXTEND THIS LIMIT.

9. *Return Policy.* Except as otherwise provided, TI does not offer any refunds, returns, or exchanges. Furthermore, no return of EVM(s) will be accepted if the package has been opened and no return of the EVM(s) will be accepted if they are damaged or otherwise not in a resalable condition. If User feels it has been incorrectly charged for the EVM(s) it ordered or that delivery violates the applicable order, User should contact TI. All refunds will be made in full within thirty (30) working days from the return of the components(s), excluding any postage or packaging costs.

10. *Governing Law:* These terms and conditions shall be governed by and interpreted in accordance with the laws of the State of Texas, without reference to conflict-of-laws principles. User agrees that non-exclusive jurisdiction for any dispute arising out of or relating to these terms and conditions lies within courts located in the State of Texas and consents to venue in Dallas County, Texas. Notwithstanding the foregoing, any judgment may be enforced in any United States or foreign court, and TI may seek injunctive relief in any United States or foreign court.

6. **免责声明:**

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